

Beyond Case 4

5.1 A Continuum of Applications

The previous chapter presented four cases. Other configurations, like inverting and noninverting gain with no offset, also are covered in previous chapters. While these constitute the vast majority of applications, some additional applications must be addressed. This is best shown in [Table 5.1](#), which lists what has been covered thus far and gives section numbers, not in parentheses, for cases addressed in this chapter. For the sake of brevity, cases with a negative reference are not covered; very few negative voltage references are manufactured or utilized. Also, for the sake of brevity, the equations are not derived, as they were in Chapter 4. They all were developed, however, using the same voltage divider, superposition, and other laws covered in previous chapters.

Table 5.1: The Gain and Offset Matrix

		Offset < 0	Offset = 0	Offset > 0
Noninverting	Gain > 1	Case 2 (4.3.2)	Noninverting gain (3.2)	Case 1 (4.3.1)
	Gain = 1	5.4	Noninverting buffer	
	Gain < 1		5.2	5.3
	Gain = 0	Voltage references		
Inverting	Gain < -1	5.7	5.5	5.6
	Gain $\geq$ -1	Case 4 (4.3.4)	Inverting gain (3.3)	Case 3 (4.3.3)

Clearly, some work is yet to be done, particularly in the cases of attenuation. For designers to have a complete set of tools to understand every combination of gain and offset that can come across their path, they need to know more than the applications with which they are most familiar (inverting and noninverting gain and noninverting buffers). The four cases presented in Chapter 4 supplemented the familiar basic circuits, and this chapter presents the rest of the cases. Voltage references, however, are not the subject of this volume.

5.2 Noninverting Attenuator with Zero Offset

The simplest of all the new cases is that of noninverting attenuation ([Figure 5.1](#)). It is done by building on the voltage divider principle (Section 2.3) and adding a unity gain op amp buffer.

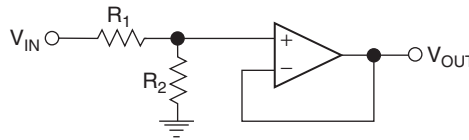


Figure 5.1: Noninverting attenuator: $V_{OUT} = mV_{IN}$, $m = R_2/(R_1 + R_2)$.

5.3 Noninverting Attenuation with Positive Offset

The case of noninverting attenuation with positive offset ([Figure 5.2](#)) is a minor variation of the noninverting attenuator, in this case, adding a second input for the reference, which is also attenuated by the voltage divider law.

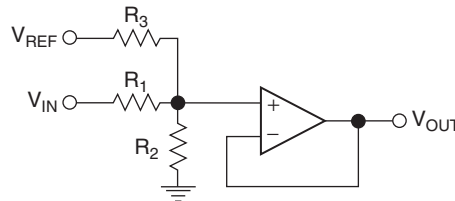


Figure 5.2: Noninverting attenuation with positive offset: $V_{OUT} = mV_{IN} + b$, $m = (1/R_1)/(1/R_1 + 1/R_2 + 1/R_3)$, $b = V_{REF} \times (1/R_3)/(1/R_1 + 1/R_2 + 1/R_3)$.

5.4 Noninverting Attenuation with Negative Offset

The case of noninverting attenuation with negative offset (Figure 5.3) is another slight variation of the noninverting buffer. In this case, instead of applying the reference by superposition to the noninverting input, the reference is applied to the inverting input through an inverting gain stage. The only limitation is that the gain on the reference must be equal to or greater than the stable bandwidth of the op amp.

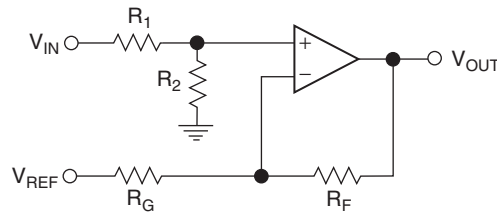


Figure 5.3: Noninverting attenuation with negative offset: $V_{OUT} = mV_{IN} - b$,
 $m = [R_2/(R_1 + R_2)] \times [1/(R_F/R_G)]$, $b = V_{REF} \times (R_F/R_G)$.

5.5 Inverting Attenuation with Zero Offset

This is the most often misdesigned circuit of all the cases. Many inexperienced designers create an unstable stage by attempting to extrapolate an inverting gain stage to the attenuation case by making R_G greater than R_F . The easiest fix for this problem is to use a voltage divider followed by a unity gain buffer, as described in Section 5.2. If inverting gain is absolutely required, then a similar solution can be implemented by adding a voltage divider to the input of an inverting gain stage.

The balance here is simple enough to understand, see Figure 5.4. R_{IN} is split into R_{INA} and R_{INB} , the sum of which cannot be greater than R_F . By the addition of R_{ATTEN} , the effective attenuation of the stage can be any value desired, while the gain of the stage from R_{INB} and R_F is always between a gain of 1 and 2.

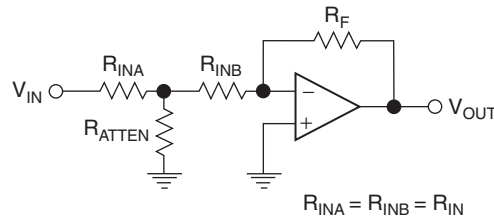


Figure 5.4: Inverting attenuation with zero offset: $V_{OUT} = mV_{IN}$,
 $m = (R_F \times R_{ATTEN})/[R_{IN} \times (R_{IN} + R_{ATTEN})]$.

5.6 Inverting Attenuation with Positive Offset

If positive offset is needed along with inverting attenuation, a combination of the previous section and [Section 5.2](#) can be employed ([Figure 5.5](#)). Just remember that the offset is also attenuated by the same factor as V_{IN} .

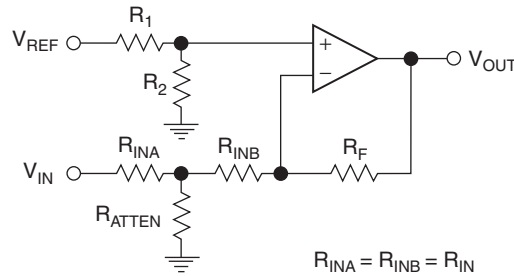


Figure 5.5: Inverting attenuation with positive offset: $V_{OUT} = mV_{IN} + b$,
 $m = (R_F \times R_{ATTEN})/[R_{IN} \times (R_{IN} + 2R_{ATTEN})]$, $b = V_{REF} \times [R_2/(R_1 + R_2)] \times [1 + R_F/(R_{IN} + R_{IN} \parallel R_{ATTEN})]$.

5.7 Inverting Attenuation with Negative Offset

If negative offset is needed along with inverting attenuation, the reference can be added to the inverting input using a voltage summation method at the inverting input ([Figure 5.6](#)). Again, it is important to have a gain of more than 1 on the V_{REF} gain channel or instability may result.

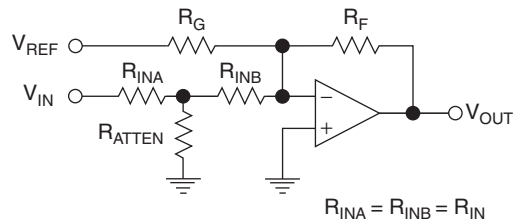


Figure 5.6: Inverting unity gain with negative offset: $V_{OUT} = mV_{IN} - b$,
 $m = (R_F \times R_{ATTEN})/[R_{IN} \times (R_{IN} + 2R_{ATTEN})]$, $b = V_{REF} \times R_F/R_G$.

5.8 Conclusion

The preceding circuits should provide a way to design just about any interface circuit the designer needs. The primary use of these gain and offset circuits, of course, is the interface between an input voltage and a data converter. The designer may also need to include filtering in their interface circuit, which is covered in later chapters.